

# DuilioBalsamo



## contacts

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LinkedIn

## languages

italian  
english  
french

## programming

Python  
Pandas  
PyTorch  
LangChain  
NLTK  
spaCy  
Scikit-learn  
SHAP  
NetworkX  
Apache Spark  
MongoDB  
GAML

born in Turin, Italy  
12 nov 1990

## Summary

I am a computational social scientist with a background in complex systems and data science. My research focuses on understanding how interactions in digitally mediated environments shape health-related behaviours and decisions — from public health monitoring and online communities, to trust and vaccination attitudes, to human interactions with AI systems. By combining computational methods, behavioural research, and experience with large-scale digital data, I aim to produce empirical knowledge capable of informing decision-making in complex sociotechnical systems.

## Research activity

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| 2022 – now  | <b>Bocconi University</b> - POSTDOCTORAL RESEARCHER<br>DONDENA Centre for Research on Social Dynamics and Public Policy<br>BIDSa Bocconi Institute for Data Science and Analytics<br>HEAL Health Emergencies Analytics Lab<br>Postdoctoral Researcher on the ERC Consolidator Grant IMMUNE: Modelling the Impact of Human Behaviours on Infection Spread. My research focuses on understanding individual and collective behaviour in the context of infectious disease spread, using machine learning, network science, and natural language processing applied to survey data, digital trace data, and social media. Within this project, I have taken a leading role in survey design and coordination of an extensive cross-national survey on vaccination behaviour, information-seeking patterns, trust, and public opinions, developing experience with the practical challenges of integrating sensitive and heterogeneous data sources. More recently, I have extended this work toward understanding trust formation in AI-generated health information. In parallel, I have engaged with EU regulatory frameworks governing data and AI — including GDPR, DSA, and the AI Act — both through research practice and through contributions to professional training in AI compliance | Milan, Italy  |
| 2019 – 2021 | <b>University of Turin, ISI Foundation</b> - PH.D. CANDIDATE<br>Ph.D. in Modeling and Data Science<br>Mathematics Dept. & Computer Science Dept. - UniTo<br>Applied Data Science Group - ISI Foundation<br>My research focused on applying computational methods to Reddit data to study complex social dynamics among individuals experiencing opioid addiction who share their experiences online. Leveraging tools from computational social science and natural language processing, I investigated public health dimensions such as opioid adoption patterns and routes of administration, as well as relational aspects including peer support mechanisms and recovery trajectories within online communities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Turin, Italy  |
| 2018        | <b>ISI Foundation</b> - JUNIOR RESEARCHER<br>Lagrange Applied Research Scholarship<br>Project supported by the Lagrange Applied Research Scholarship, focusing on text mining and machine learning in the field of public health monitoring, on data gathered from Reddit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Turin, Italy  |
| 2017        | <b>ISI Global Science Foundation</b> - JUNIOR RESEARCHER<br>Applied Data Science Group<br>Investigating the emergence of anomalies and the evolution of time resolved topics in large news and social media document collections: GDELT, Twitter and Reddit. Project in collaboration with DensityDesign Lab of Politecnico di Milano.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | New York, USA |
| 2016        | <b>aizoOn Technology Consulting</b> - INTERN<br>Research & innovation Dept.<br>Internship finalised at the development of a Proof of Concept and a Master's Thesis. I studied the come out of emergency hard brakings of high speed trains, using data from train's black boxes and clustering the driving behaviours of several sample trains.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Turin, Italy  |

## Fellowships and awards

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|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| 2022 – now  | <b>BIDSA Bocconi Institute for Data Science and Analytics</b><br>Fellowship                                                                                                                                                                                          | Bocconi University, Italy                      |
|             | <b>DONDENA Centre for Research on Social Dynamics and Public Policy</b><br>Fellowship                                                                                                                                                                                | Bocconi University, Italy                      |
|             | <b>HEAL - Health Emergencies Analytics Lab (former Covid Crisis Lab)</b><br>Fellowship                                                                                                                                                                               | Bocconi University, Italy                      |
| 2019 – 2021 | <b>Ph.D. Student Fellowship</b><br>Student Fellowship<br>Fully funded national Ph.D. scholarship                                                                                                                                                                     | University of Turin, Italy                     |
| 2017        | <b>Lagrange Applied Research Scholarship 2017</b><br><br>Scholarship<br>Funded by Fondazione CRT, scholarship supporting projects focusing on Data Science and Digital epidemiology with a multidisciplinary approach between science, technology and public health. | Fondazione CRT, Italy<br>ISI Foundation, Italy |

## Education

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| 2019 – 2021 | <b>Ph.D. - MODELING AND DATA SCIENCE</b><br><br>Mathematics Dept. & Computer Science Dept. - UniTo<br>Applied Data Science Group - ISI Foundation<br>Thesis: "The opioid epidemic through the lens of social media: mining digital traces for public health, pharmacovigilance, and rehabilitation."<br>Supervisors: <ul style="list-style-type: none"><li>• Rossano Schifanella, Computer Science Dept. UniTo</li><li>• Paolo Bajardi, ISI Foundation.</li></ul> | University of Turin, Italy<br>ISI Foundation, Italy                                  |
| 2013 – 2016 | <b>Master's degree - PHYSICS OF COMPLEX SYSTEMS</b><br>Physics Dept.<br>Thesis: "Emerging patterns and clustering driving behaviours for a data-driven perspective in railways transportation systems."<br>Supervisors: <ul style="list-style-type: none"><li>• Ciro Cattuto, ISI Foundation</li><li>• Paolo Bajardi, AizOon Technology consulting.</li></ul>                                                                                                     | 110/110 University of Turin, Italy                                                   |
| 2011 – 2012 | <b>Erasmus Program - PHYSICS</b><br><br>Physics Dept.                                                                                                                                                                                                                                                                                                                                                                                                             | University Claude Bernard Lyon 1, France<br>École normale supérieure de Lyon, France |
| 2009 – 2013 | <b>Bachelor's degree - PHYSICS</b><br>Physics Dept.<br>Thesis: "Measurement of the muonic component in ultra high energy extensive air showers"                                                                                                                                                                                                                                                                                                                   | 99/110 University of Turin, Italy                                                    |

## Certificates

|      |                                                                                                                           |                               |
|------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 2017 | <b>Machine Learning Foundations: A Case Study Approach</b><br>Coursera Course Certificates<br>Certificate n. AN5AS2VNZETK | University of Washington, USA |
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# Publications

Notes - The publications most relevant for the AI2HEALTH position are highlighted with \*. My publication record reflects two periods of parental leave (2021 and 2023), which I flag solely to contextualise the pace of output over the past five years

- 2026      **Implications for distancing measures on in-person school and work attendance from Italian post-pandemic social contact data**  
LUCCHINI L., MARZIANO V., TRENTINI F., CHIAVENNA C., D'AGNESE E., OFFEDDU V., MANICA M., POLETTI P., BALSAMO D., GUZZETTA G., AJELLI M., MELEGARO A., MERLER S.  
Communications Medicine, 2026  
<https://www.nature.com/articles/s43856-026-01543-4>
- 2025      **Mapping inequities in global vaccine sentiment research \***  
BALSAMO D., OFFEDDU V., AGHAMOHAMMADI Z., CHIAVENNA C., LEONE L. P., D'AGNESE E., SHARMA D., TORBICA A., SHAYEGH S., ANDREU-PEREZ J., MELEGARO A.  
BMJ Global Health, 10(11), 2025  
<https://gh.bmj.com/content/10/11/e018587>
- Advancing coupled behavioural-epidemic models: An interdisciplinary approach to integrate behavioural drivers into epidemic spread and vaccination decisions**  
OFFEDDU V., COLOSI E., LUCCHINI L., LEONE L. P., CHIAVENNA C., BALSAMO D., D'AGNESE E., BONACINA F., CUCCINIELLO M., TRENTINI F., ALETA A., MANFREDI P., MORENO Y., KARSAI M., COLIZZA V., KOLTAI J., MELEGARO A.  
Preprint medRxiv, 2025 - under review  
<https://www.medrxiv.org/content/10.1101/2025.10.07.25337410v1>
- 2023      **The pursuit of peer support for opioid use recovery on Reddit \***  
D. BALSAMO, P. BAJARDI, G. DE FRANCISCI MORALES, C. MONTI, R. SCHIFANELLA  
In Proceedings of the International AAAI Conference on Web and Social Media, 2023  
<https://doi.org/10.1609/icwsm.v17i1.22122>
- 2021      **Patterns of Routes of Administration and Drug Tampering for Nonmedical Opioid Consumption: Data Mining and Content Analysis of Reddit Discussions**  
D. BALSAMO, P. BAJARDI, A. SALOMONE, R. SCHIFANELLA  
In Journal of Medical Internet Research 23 (1), e21212, 2021  
<https://www.jmir.org/2021/1/e21212/>
- 2019      **Firsthand Opiates Abuse on Social Media: Monitoring Geospatial Patterns of Interest Through a Digital Cohort \***  
D. BALSAMO, P. BAJARDI, A. PANISSON  
The World Wide Web Conference (pp. 2572-2579). ACM, 2019  
<https://doi.org/10.1145/3308558.3313634>
- Inside the Echo Chamber: Disentangling network dynamics from polarization**  
D. BALSAMO, V. GELARDI, C. HAN, D. RAMA, A. SAMANTRAY, C. ZUCCA, M. STARNINI  
Preprint arXiv, 2019  
<https://arxiv.org/abs/1906.09076>
- Topic Tomographies (TopTom): a visual approach to distil information from media streams**  
B. GOBBO, D. BALSAMO, M. MAURI, P. BAJARDI, A. PANISSON, P. CIUCCARELLI  
Computer Graphics Forum (Vol. 38, No. 3, pp. 609-621), 2019  
<https://doi.org/10.1111/cgf.13714>

## In preparation

### **The multi-dimensional role of trust and peer influence in vaccination opinions \***

BALSAMO D., OFFEDDU V., CHIAVENNA C., LUCCHINI L., LEONE L. P., COLOSI E., BONACINA F., D'AGNESE E., MELEGARO A.

### **Trust and authority signals in COVID-19 vaccination content on TikTok \***

FALLARA D., MEJOVA Y., BALSAMO D.

### **Vaccination decisions and exposure to TikTok and YouTube: evidence from linked survey and digital trace data**

BALSAMO D., MURATORI D., MELEGARO A.

## Teaching activity

- 2026 **Fondamenti Tecnici dell'Intelligenza Artificiale** - INSTRUCTOR Rome, Italy  
Academy Istituto Privacy - Istituto Italiano per la Privacy  
Delivered lecture on "Technical Foundations of Artificial Intelligence" in the context of the AI Act, as part of the Advanced Course "Le mille facce dell'Intelligenza Artificiale"
- 2025 – now **Simulation and Modeling** - INSTRUCTOR Milan, Italy  
MSc in Data Science and Business Analytics, Bocconi University  
Delivered lectures on agent-based modeling, network theory, and spreading phenomena on networks. Designed and coordinated hands-on coding laboratories and Jupyter notebook-based exercises in Python. Developed examination materials and was responsible for grading and assessment.
- 2025 – now **Foundations of Social Sciences - Module I (Empirical Research Methods and Data Analysis)** - INSTRUCTOR Milan, Italy  
MSc in Economic and Social Sciences, Bocconi University  
Delivered lectures on computational social science methods, including digital data collection (scraping and APIs), supervised and unsupervised machine learning, text-as-data approaches, large language models, and social network analysis. Designed and coordinated hands-on coding laboratories and Jupyter notebook-based exercises in Python. Developed examination materials and assignments, supervised student projects, and was responsible for grading and assessment.
- 2022 – 2025 **Simulation and modeling** - TEACHING ASSISTANT Milan, Italy  
MSc in Data Science and Business Analytics, Bocconi University  
Giving front lectures on network theory and spreading phenomena on networks. Developing and teaching practical coding. Writing and correcting exams.
- 2024 **Sociology** - INVITED LECTURER Milan, Italy  
BSc in Economics, Management and Computer Science, Bocconi University  
Invited lecture: "The Opioid Epidemic Through the Lens of Reddit - A Computational Social Science Approach"
- 2022 **Sociology** - INVITED LECTURER Milan, Italy  
BSc in International Politics and Government, Bocconi University  
Invited lecture: "Covid-19 Vaccine Hesitancy on Reddit - A Computational Social Science approach"
- PhD program lectures** - INVITED LECTURER Milan, Italy  
Department of Social and Political Sciences, Bocconi University  
Invited lecture: "Modelling the impact of human behaviours on infections spread - A Computational Social Science approach"

## Supervision

Supervision and mentoring of master's students, research fellows, and early-career researchers in computational social science and data science.

- 2026 **Supervisor** - LAGRANGE FELLOW Turin, Italy  
Largrange Scholarship program, ISI Foundation, Turin  
Co-supervisor: Yelena Mejova
- Title: "Trusting the Machine: Health information seeking and trust formation in Large Language Models "
- Student fellow: Arianna Longo

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| 2025–2026 | <b>Supervisor</b> - LAGRANGE FELLOW<br>Largrange Scholarship program, ISI Foundation, Turin<br>Co-supervisor: Yelena Mejova<br><br>Title: "Trust and authority signals in COVID-19 vaccination content on TikTok"<br>Student fellow: Davide Fallara                                                                                                                                                                                                                                                                    | Turin, Italy |
| 2025      | <b>Tutor</b> - MASTER THESIS<br>MSc in Data Science and Business Analytics, Bocconi University<br>Supervisor: Alessia Melegaro<br><br>Title: "YouTube exposure and vaccination decisions: evidence from linked survey and digital trace data"<br>Candidate: Davide Muratori<br><br>Title: "Graph Neural Network–Based Social Recommender Systems: Echo Chambers and Polarization on Reddit"<br>Candidate: Arianna Zottoli                                                                                              | Milan, Italy |
| 2024      | <b>Tutor</b> - MASTER THESIS<br>MSc in Data Science and Business Analytics, Bocconi University<br>Supervisor: Alessia Melegaro<br><br>Title: "Opinion Dynamics on Social Media: The Role of Recommendation Systems and Algorithmic Bias"<br>Candidate: Gaia Triggiani<br><br>Title: "An NLP Analysis of Vaccination Campaigns on Twitter and TikTok"<br>Candidate: Sofia Maselli<br><br>Title: "The Spread of Misinformation on Social Media during the Covid-19 Pandemic: A Reddit Case"<br>Candidate: Andrea Santoro | Milan, Italy |
| 2023–2024 | <b>Tutor</b> - RESEARCH ASSISTANT<br>IMMUNE Project, Bocconi University<br>Supervisor: Alessia Melegaro<br><br>Project: "Mapping inequities in global vaccine sentiment research"<br>RA: Zhina Agamohammadi                                                                                                                                                                                                                                                                                                            | Milan, Italy |
| 2023      | <b>Tutor</b> - MASTER THESIS<br>MSc in Data Science and Business Analytics, Bocconi University<br>Supervisor: Alessia Melegaro<br><br>Title: "Computer Vision Food Classification of Instagram User-Generated Content for Predicting Sales during COVID-19"<br>Candidate: Dominique Di Cesare                                                                                                                                                                                                                          | Milan, Italy |
| 2022      | <b>Tutor</b> - MASTER THESIS<br>MSc in Data Science and Business Analytics, Bocconi University<br>Supervisor: Alessia Melegaro<br><br>Title: "Covid-19 Vaccine Hesitancy: Network-based Analysis using NLP"<br>Candidate: Jakob Schlierf                                                                                                                                                                                                                                                                               | Milan, Italy |

# Conferences, seminars and schools

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|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 2026 | <b>NETSCI 2026: International School &amp; Conference on Network Science</b><br>Northeastern University                                                                                                                                                   | Boston, MA, USA            |
|      | <b>CS2Italy 2026: Italian Conference on Computational Social Science</b><br>ISI Foundation<br>Served as Program Committee                                                                                                                                 | Turin, Italy               |
| 2025 | <b>EPIDEMICS 10: 10th International Conference on Infectious Disease Dynamics</b><br>USA<br>Elsevier                                                                                                                                                      | San Diego, CA, USA         |
|      | <b>IC2S2 - 11th International Conference on Computational Social Science</b><br>Linköping University                                                                                                                                                      | Norrköping, Sweden         |
|      | <b>CS2Italy -- The Italian Conference on Computational Social Science</b><br>University of Trento                                                                                                                                                         | Trento, Italy              |
| 2024 | <b>Workshop We4Us 2025 - Physics of Data</b><br>University of Padua<br>Invited lecturer within the workshop.                                                                                                                                              | Padua, Italy               |
|      | <b>Workshop on Infectious Disease Modelling with Human Behaviour</b> - ORGANISER<br>Covid Crisis Lab & DONDENA Centre, Bocconi University<br>Organised an international workshop with global experts on human behaviour and infectious disease modelling. | Milan, Italy               |
|      | <b>Data Donation Symposium 2024</b><br>University of Amsterdam                                                                                                                                                                                            | Amsterdam, The Netherlands |
|      | <b>DAISI Advanced AI Methods Workshop</b><br>DONDENA Centre for Research on Social Dynamics and Public Policy<br>Bocconi University                                                                                                                       | Milan, Italy               |
|      | <b>CCS - Conference on Complex Systems 2024</b><br>Complex Systems Society                                                                                                                                                                                | Exeter, London             |
|      | <b>DAISI Dondena AI and Society Initiative Seminar</b><br>DONDENA Centre for Research on Social Dynamics and Public Policy<br>Bocconi University                                                                                                          | Milan, Italy               |
|      | <b>CIVICA Data Science Seminar</b><br>CIVICA Research                                                                                                                                                                                                     | Online                     |
| 2023 | <b>NetSciX - International School and Conference on Network Science</b><br>Network Science Society                                                                                                                                                        | Venice, Italy              |
|      | <b>Epidemics - 9th International Conference on Infectious Disease Dynamics</b><br>Elsevier                                                                                                                                                                | Bologna, Italy             |
|      | <b>IC2S2 - 9th International Conference on Computational Social Science</b><br>Technical University of Denmark                                                                                                                                            | Copenhagen, Denmark        |
|      | <b>ICWSM - The 17th International AAAI Conference on Web and Social Media</b><br>AAAI                                                                                                                                                                     | Limassol, Cyprus           |

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|      | <b>Do.Re.Mee Seminars</b><br>DONDENA Centre for Research on Social Dynamics and Public Policy<br>Bocconi University   | Milan, Italy             |
|      | <b>1st CIVICA Conference on European Polarisation</b><br>CIVICA Research                                              | Online                   |
| 2022 | <b>ELLIS@Milan Artificial Intelligence workshop</b><br>Computing Sciences Department<br>Bocconi University            | Milan, Italy             |
|      | <b>CCS - Conference on Complex Systems 2022</b><br>Complex Systems Society                                            | Palma De Mallorca, Spain |
|      | <b>CLEAN Crime Law and Economic Analysis Seminars</b><br>Bocconi University                                           | Milan, Italy             |
|      | <b>CIVICA Conference</b><br>CIVICA Research                                                                           | Budapest, Hungary        |
|      | <b>Complex Networks: Theory, Methods, and Applications</b><br>Lake Como School of Advanced Studies                    | Como, Italy              |
| 2021 | <b>IC2S2 - 7th International Conference on Computational Social Science</b><br>ETH Zurich                             | Zurich, Switzerland      |
|      | <b>CCS - Conference on Complex Systems 2021</b><br>Complex Systems Society                                            | Lyon, France             |
| 2019 | <b>Summer School Series on Methods for Computational Social Science</b><br>Gesis Leibniz Institut for Social Sciences | Berlin, Germany          |
| 2018 | <b>Complexity 72h Workshop</b><br>IMT School for Advanced Studies                                                     | Lucca, Italy             |
|      | <b>DSAA - 5th IEEE International Conference on Data Science and Advanced Analytics</b><br>IEEE                        | Turin, Italy             |